NIKOLAY KARPUSHA

Senior Unity Developer / Gameplay & Systems Lead

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PROFILE

Senior Unity Developer with 7+ years of commercial Unity experience and 12+ years in game development. Specialized in gameplay programming, architecture and complex gameplay systems. Experienced in building systems from scratch, developing existing projects, optimizing large-scale gameplay scenes and creating internal tools for game designers.

CORE EXPERTISE

GAMEPLAY & AI	ARCHITECTURE	PERFORMANCE & UNITY
Gameplay Programming	C# · OOP · SOLID	GPU Instancing · Job System
Combat Systems	Zenject / Dependency Injection	CPU / GPU Optimization
Utility AI	Modular C# Services	Unity Profiler
Tactical Systems & Formations	Factory · State · Object Pool	Unity Editor Tools
Movement · Pathfinding	ScriptableObjects	Timeline · NavMesh · FMOD
State Machines	Addressables	Git / GitLab

PROFESSIONAL EXPERIENCE

Senior Unity Developer / Gameplay & Systems Lead — Sep 2025 — Present

Team 175 • Veterans: Napoleonic Wars

- Lead gameplay and systems programmer; implemented the majority of the game's core gameplay mechanics and systems.
- Designed the core architecture around independent C# services and Zenject DI; used factories and object pooling.
- Implemented movement, formations, combat, ranged combat, morale, fatigue, artillery, charges, retreats and ammunition types.
- Developed Utility AI and tactical unit control; implemented movement and pathfinding for thousands of units.
- Implemented high-performance rendering of large numbers of units using GPU Instancing.
- Created Unity tools for balancing, gameplay testing and content production; developed a custom game-time system with active pause.
- Integrated FMOD, Addressables and Unity Input System.

Gameplay & Systems Programmer — In Development

Team 175 • New IP — Samurai Project

- Built the technical foundation using architecture and systems developed for Veterans; implemented DI and core services.
- Implemented the foundation of the combat system and related gameplay mechanics.
- Designed and implemented an Orchestrator system for defining and automatically executing sequences of gameplay actions.
- Used the Orchestrator to create gameplay scenes, events, character movement, enemy spawning and combat sequences without separate custom logic for each scenario.
- Built the project's first cinematic sequence using the Orchestrator.

Senior Unity Developer — Jul 2025 — Sep 2025

Samustai • Legacy Unity Project Recovery

- Recovered two Unity projects after loss of .meta files and broken asset references.
- Restored scenes, prefabs, gameplay and combat systems, materials, VFX and system dependencies.
- Fixed critical launch issues and prepared projects for handover to new development teams.

Unity C# Developer — Apr 2020 — Jan 2025

Game Labs • Ultimate Admiral: Dreadnoughts

- Developed a modular ship constructor, global strategic map and ballistic system; reworked shipbuilding gameplay and key UI elements.
- Implemented localization into 9 languages, Steam API integration and MessagePack serialization.
- Created Unity Editor Tools for game designers; participated in prototyping and technical evaluation of new gameplay concepts.

PROFESSIONAL EXPERIENCE — CONTINUED

Unity C# Developer — Aug 2019 — Mar 2020

NoDeadLineGames • Magnia

- Gameplay systems, FMOD and UI; Steam Achievements and performance optimization.
- Localization into 28 languages; created tools and scripts for 3D/VFX artists.

Unity C# Developer — Nov 2017 — Aug 2019

OMG Studio • Multiple Projects

- Physia — gameplay levels, FMOD, GUI, lighting and mobile optimization.
- Max & Maya — VFX, water and underwater environments, animation and sound.
- Crash KnockDown — FMOD and UI; Gorilla Action / Gorilla Idle Game — AI, UI, audio and team coordination.
- Participated in Physia development during a business trip to China.

Game Developer — Jan 2014 — Nov 2017

RedLegGames • Indie Game Development

- Full-cycle indie game development from concept to release: gameplay design and programming, art and audio integration, multi-platform porting, trailers and game content.

RELEASED / RELEVANT PROJECTS

Veterans: Napoleonic Wars	RTS / Tactical Strategy
Ultimate Admiral: Dreadnoughts	Strategy
Lost Jelly	Game / Project
Magnia	Puzzle
Road of Dust and Rust	Action / Adventure
Swords And Their Boy	Action / Adventure

EDUCATION

Novosibirsk State Technical University — Faculty of Mechanical Engineering, Mechanical Engineering Technology.
Akadempark / Alawar Winter School — Game Development.
Universarium / Wargaming — Game Design.

LEADERSHIP & ADDITIONAL

- Technical leadership, ownership of gameplay systems, task decomposition and collaboration with Game Designers, Artists and VFX.
- GitHub: github.com/DjKarp • Portfolio: redleggames.com
- References: Alexander Petrov, Maxim Zasov — Game Labs Producers.